FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – III
PAPER –2
TEST DATE: 25-12-2022

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 – 04, 19 – 22, 37 – 40)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A *(05 –10, 23 – 28, 41 – 46)*: This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B *(11 – 18, 29 – 36, 47 – 54)*: This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.**

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

Physics

PART – I

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

1. The refracting angle of a prism is 60º. What should be the minimum value of the refractive index of the material from which the prism is to be made so that light cannot emerge from the second refracting surface.

(A) $\frac{\sqrt{3}}{2}$ (B) 2

(C) 1.5 (D) $\frac{2}{\sqrt{3}}$

2. The de-Broglie wavelength of Helium atom of mass 'm' having energy corresponding to the mean thermal energy at a temperature of 'T' Kelvin (k_B is Boltzmann constant) is

(A) $\frac{h}{\sqrt{6mk_BT}}$ (B) $\frac{h}{\sqrt{3mk_BT}}$

(C) $\frac{h}{\sqrt{2mk_BT}}$ (D) $\frac{h}{\sqrt{mk_BT}}$.

3. If K_α radiation of copper Cu(Z=29) has a wavelength of 1.52 Å, the wavelength of the corresponding radiation of molybdenum Mo($Z = 42$) is

(A) 0.71 Å (B) 1.71 Å

(C) 0.50 Å (D) 1.5 Å.

4. The atomic ratio between the uranium isotopes ^{238}U and ^{234}U in a mineral is 1.8×10^4. The activity of both the isotopes at the given moment is same. The half life of ^{238}U is $T_{1/2}(^{238}U) = 4.5\times10^9$ yr, the half life of ^{234}U is

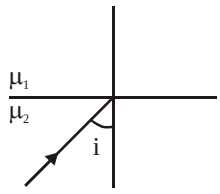
(A) 4.05×10^{13} yr (B) 2.5×10^5 yr

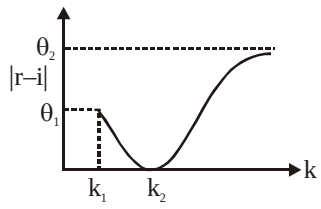
(C) 2.00×10^{13} yr (D) 8.1×10^{13} yr.

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

5. The figure shown a ray of light incident at an angle i = π/3. For different values of μ_1 and μ_2 angle of refraction is difference. The plot drawn shows the variation of |r – i| versus $\frac{\mu_1}{\mu_2} = k$, (r = angle of refraction).



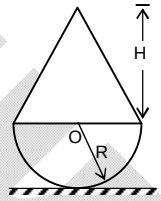


(A) The value of k_1 is $\frac{2}{\sqrt{3}}$ (B) The value of $\theta_1 = \pi/6$

(C) The value of $\theta_2 = \pi/3$ (D) The value of k_2 is 1

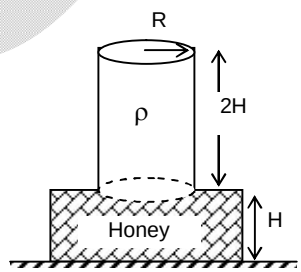
6. Two plane mirrors at an angle such that a ray incident on a mirror undergoes a total deviation of 240º after two reflections.
(A) The angle between the mirrors is 60º.
(B) The number of images formed by this system will be 5, if an object is placed symmetrically between the mirrors.
(C) The number of images will be 5 if an object is kept un-symmetrically between the mirrors.
(D) A ray will retrace its path after 2 successive reflections, if the angle of incidence with one mirror is 60º.

7. It is desired to construct a toy by attaching a solid cone of height H on the top of solid hemisphere as shown in the figure. The radius of the hemisphere and the base of cone is R. The whole system is kept on a rough horizontal surface. If the toy is slightly disturbed from the given equilibrium position it oscillates simple harmonically. Then mark the correct statements. (Given that both the cone and hemisphere are made of same material)



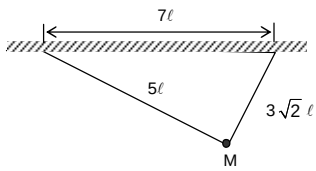
(A) $H < \sqrt{3}R$
(B) $H > \sqrt{3}R$
(C) $\sqrt{3}R < H < 2R$
(D) centre of mass of the combined system must not be above O.

8. A bottle is kept on the ground as shown in the figure. The bottle can be modelled as having two cylindrical zones. The lower zone of the bottle has a cross-sectional radius of $R\sqrt{2}$ and is filled with honey of density 2ρ. The upper zone of the bottle is filled with the water of density ρ and has a cross-sectional radius R. The height of the lower zone is H while that of the upper zone is 2H. If now the honey and the water parts are mixed together to form a homogeneous solution (Assume that total volume does not change)



(A) The pressure inside the bottle at the base will remain unaltered.
(B) The normal reaction on the bottle from the ground will remain unaltered.
(C) The pressure inside the bottle at the base will increase by an amount $(1/2)\rho gH$
(D) The pressure inside the bottle at the base will decrease by an amount $(1/4)\rho gH$

9. A particle is suspended by two ideal strings as shown in the figure. Now mass m is given a small displacement perpendicular to the plane of triangle formed. Choose the correct statement(s).



(A) The period of oscillation of the system is $2\pi\sqrt{\frac{3\sqrt{3}\ell}{g}}$.
(B) The period of oscillation of the system is $2\pi\sqrt{\frac{3\ell}{g}}$.
(C) The period of oscillation of the system is independent of M.
(D) If the distance between the suspension points was kept constant and the lengths of the strings were quadrupled then the period of the system will double.

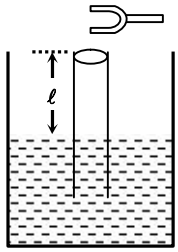
10. Two spherical bodies, each of mass M, are kept with their centres at a distance 2L apart. A particle of mass m is projected from the mid-point of the line joining their centres, perpendicular to the line. The gravitational constant is G. Which of the following statements is/are true?
(A) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $4\sqrt{\frac{GM}{L}}$
(B) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $2\sqrt{\frac{GM}{L}}$
(C) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $\sqrt{\frac{2GM}{L}}$
(D) The total mechanical energy of the mass m remains constant.

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

11. A man is looking down a 9 m deep tank filled with a transparent liquid of refractive index $\sqrt{3}$. The line of sight of the man makes an angle of 30° with the horizontal. What is the apparent depth of the tank (in meter) as seen by the man?

12. A tuning fork of known frequency is held at the open end of a long tube, which is dipped into water as shown in the figure. The tuning fork of frequency 165 Hz resonates in first and second harmonic when air columns have lengths ℓ_1 = 50 $\pm$0.5 cm and ℓ_2 = 150 $\pm$ 0.1 cm. Find the maximum possible error in measurement of speed of sound(in m/s) in air.



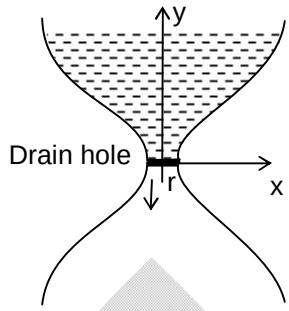
13. A standing wave exists in a string of length 150cm and is fixed at both ends. The displacement amplitude of a point at a distance of 10cm from one of the ends is $5\sqrt{3}$ mm. The distance between the two nearest points, within the same loop and having displacement amplitude equal to $5\sqrt{3}$ mm, is 10cm. Find the mode of vibration of the string i.e. the harmonic produced.

14. A uniform inextensible string of length L= 6m and linear density μ is suspended vertically and a transverse pulse is produced at the lower end. At the same time a point object is released from rest from the top of the string and falls freely. How far (in meter) from the top does the object passes the pulse?

15. A person standing between two parallel hills, fires a gun. He hears the first echo after 1.5sec and the second after 2.5 sec. The speed of sound is 332 m/s. When will he hear the third echo?

16. A point source of power 50π watts is producing sound waves. The velocity of sound is 330m/s, density of air is 1.0 kgm^{-3}. Then find the pressure amplitude (in Nm^{-2}) at a distance r = $\sqrt{330}$m from the point source.

17. The shape of an ancient water clock jar is such that the water level descends at a constant rate at all times. The water level falls by λ m/s and the shape of the jar is given by $y \alpha x^n$. Find the value of n if the radius r of the drain hole can be assumed to be very small.



18. During Searle's experiment, the zero of a vernier scale lies between 3.20×10^{-2} m and 3.25×10^{-2} m of the main scale divisions. The 20th division of the vernier scale exactly coincides with one of the main scale divisions. When an additional load of 2 kg is applied to the wire, the zero of the vernier scale still lies between 3.20×10^{-2} m and 3.25×10^{-2} m of the main scale but now the 45th division of the vernier scale coincides with one of the main scale divisions. The length of the thin metallic wire is 2 m and its cross-sectional area is 8.0×10^{-7} m^2. The least count of the vernier scale is 1.0×10^{-5} m. Find the maximum percentage error in the Young modulus of the wire.

Chemistry

PART – II

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

19. The colourless inter halogen compound among the following is
(A) IF_3
(B) IBr
(C) ICl
(D) IF_7

20. What will be the value of entropy of mixing for 44 gm CO_2 gas, 28 g N_2 gas and 2 mole of Ar gas at 400 K. Considering ideal behaviour of gases.
(A) $6R\ell n\frac{1}{2}$
(B) $\frac{3}{200}\ell n2$
(C) $6R\ell n2$
(D) $2400R\ell n2$

21. If the concentration (C) of a strong electrolyte is diluted 16 times. The magnitude of change in molar conductivity will be? (b is Debye, Huckel Onsagar constant)
(A) $\frac{4}{3}b\sqrt{C}$
(B) $\frac{b}{4}\sqrt{C}$
(C) 0
(D) $\frac{3}{4}b\sqrt{C}$

22. Which of the following is correct electronic configuration of $_{64}Gd^{2+}$ ion?
(A) $[Xe]4f^8 5d^o 6s^2$
(B) $[Xe]4f^7 5d^o 6s^1$
(C) $[Xe]4f^5 5d^1 6s^2$
(D) $[Xe]4f^7 5d^1 6s^0$

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

23. The correct option(s) is/are
(A) In 1 equivalent of H_2SO_4, 2 equivalent of H-atom is present.
(B) Volume strength of $\frac{1}{5}$N H_2O_2 solution is 1.12.
(C) 14 gm of ethene required, 48 gm of oxygen gas for complete combustion.
(D) Number of moles of oxygen atom in 276 gm of N_2O_4 is 6.

24. Which of the following solution(s) is/are isotonic with 0.2 M NaCl $(\alpha = 1)$ solution at identical temperature. (α = degree of dissociation)
(A) 0.2 M $MgCl_2$ $(\alpha = 0.5)$ solution.
(B) 18 g of glucose present in 250 ml solution.
(C) 0.5 mole of $Al(NO_3)_3$ $(\alpha = 0.4)$ present in 4 litre solution.
(D) 1 mole $AlCl_3$ $(\alpha = 0.2)$ present in 4 litre solution.

25. The C – O bond length in $BH_3.CO$, $\left[Mn(CO)_6\right]^+$, $\left[V(CO)_6\right]^-$ and $\left[Ti(CO)_6\right]^{2-}$ are w, x, y and z $\overset{o}{A}$ respectively. The correct option is/are
(A) x > w
(B) y < z
(C) z < x
(D) y > w

26. For 1 mole of an ideal gas
$\left(\frac{dP}{dT}\right)_V \cdot \left(\frac{dV}{dT}\right)_P =$
(A) $\frac{R}{T}$ (B) $\frac{1}{T}$
(C) $\frac{R^2}{PV}$ (D) $\frac{R^2}{P^2}$

27. Which of the following statement(s) is/are true about PH_3?
(A) It is a toxic gas.
(B) It can be prepared by hydrolysis of white phosphorous with NaOH solution.
(C) It is more basic than NH_3.
(D) The P – H bond in PH_3 is stronger than N – H bond in NH_3.

28. The correct statement(s) is/are
(A) NH_4^+ and Mg^{2+} cations (basic radical) belongs to zero group.
(B) Chloride of Hg(I) and Hg(II) do not respond to chromyl chloride test.
(C) In the brown ring test of NO_2^-, the $FeSO_4$ solution must be freshly prepared.
(D) Hg^{+2} ion on reaction with excess KI produce yellow coloured HgI_2 precipitate.

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

29. How many of the following cations (basic radicals) do not belong to group-V.
Ni^{+2}, Mg^{+2}, Sr^{+2}, Zn^{+2}, K^+, Fe^{+3}, Ca^{+2}, Mn^{+2}, Pb^{+2}

30. The number of white coloured compounds among the following
Ag_2CO_3, Ag_2SO_3, $AgNO_2$, Ag_2S, AgI, Ag_2CrO_4, PbS, ZnS, FeS, $Fe(OH)_3$, $BaCO_3$.

31. In crystal structure of Fe_3O_4 metal ions occupy void (Tetrahedral and octahedral) space. While O^{2-} ion present at FCC lattice. What will be total number of unpaired electron present in metal ion occupying tetrahedral voids per unit cell.

32. How many of the following ores contains CO_3^{2-} ion.
Cerrusite, Azurite, Calamine, Zincite, Siderite, Magnetite, Magnesite, Dolomite, Bauxite.

33. How many of the following species are coloured due to charge transfer
$\left[Co(H_2O)_6\right]^{2+}$, $Fe_4\left[Fe(CN)_6\right]_3$, V_2O_5, $\left[Ti(H_2O)_6\right]^{3+}$, $K_2Cr_2O_7$, MnO_4^-, $\left[Mn(H_2O)_6\right]^{2+}$

34. Total number of correct statements among the following is
1. Pressure and density are intensive properties.
2. Every isolated system is closed system and vice-versa.
3. Entropy change for adiabatic reversible process is zero.
4. A reaction is non-spontaneous at any temperature when enthalpy change is positive and entropy change is negative.
5. For cyclic process internal energy change is zero.
6. For isobaric process, volume (for fixed amount of an ideal gas) get doubled when temperature increases from 100^oC to 200^oC.
7. The dissolution of NaCl in H_2O is endothermic though it is spontaneous process.
8. Heat of combustion of ethane is – 341.1 kcal/mole and heat of combustion of ethyne is – 310 kcal/mole means ethane is a better fuel.

35. Total number of geometrical isomers for complex
$\left[Fe(en)(F)(Cl)(Br)(I)\right]^{2-}$ is (en = 1, 2-ethylenediamine).

36. How many of the following Antibiotics has inhibitory effect on microbes
Peniciline, Ertythromycin, Ofloxacin, Tetracyclin, Amonoglucosides, Chloramphenical.

Mathematics

PART – III

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

37. A real number a is chosen randomly and uniformly from the interval [–20, 18] the probability that the roots of the polynomial $x^4 + 2ax^3 + (2a - 2)x^2 + (-4a + 3)x - 2$ are all real is

(A) $\frac{17}{19}$ (B) $\frac{13}{19}$

(C) $\frac{21}{19}$ (D) $\frac{18}{19}$

38. Find the number of permutations x_1, x_2, x_3, x_4, x_5 of numbers 1, 2, 3, 4, 5 such that the sum of five products $x_1x_2x_3 + x_2x_3x_4 + x_3x_4x_5 + x_4x_5x_1 + x_5x_1x_2$ is divisible by 3

(A) 16 (B) 80

(C) 32 (D) 48

39. In the complex plane, let A be the set of solutions to $z^3 - 8 = 0$ and let B be the set of solution to $z^3 - 8z^2 - 8z + 64 = 0$, then the maximum distance between a point of A and a point of B is

(A) $\sqrt{21}$ (B) $3\sqrt{21}$

(C) $4\sqrt{21}$ (D) $2\sqrt{21}$

40. Let $f(x) = \{x\}[x]$; $g(x) = ax^2$ the sum of all real solutions of equation satisfying $f(x) = g(x)$ is 420. (Where a is (+ve) rational number), then a is equal to
(Where [.] and {.} represents greatest integer function and fractional part function respectively)

(A) $\frac{28}{900}$ (B) $\frac{29}{900}$

(C) $\frac{31}{900}$ (D) $\frac{37}{900}$

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

41. Let $P(x) = ax^2 + bx + c$ where a, b, c are real numbers and $a \neq 0$, the polynomial $P(x^2 + 4x - 7)$ has x = 1 as a root and at least one repeated roots, then the roots of polynomial $P(x^2 + 4x - 7)$ can be

(A) x = 1 (B) x = –5

(C) x = –2 (D) x = 2

42. If $b_n = \sum_{r=0}^{n} \frac{1}{{}^nC_r}$, $n \geq 1$, then b_n can be written as

(A) $b_n = \frac{1}{n} \cdot \sum_{r=0}^{n-1} \frac{r+1}{{}^{n-1}C_r} + 1$ (B) $b_n = \frac{1}{n} \cdot \sum_{r=0}^{n-1} \frac{r+1}{{}^{n-1}C_r}$

(C) $b_n = \frac{n+1}{2n} \cdot b_{n-1} + 1$ (D) $b_n = \frac{n-1}{2n} \cdot b_{n-1} + 1$

43. Equation of two lines passes through point A and B whose position vectors are $\vec{a}$ and $\vec{b}$ is $\vec{r} = \vec{a} + \lambda\vec{\alpha}$ and $\vec{r} = \vec{b} + \mu\vec{\beta}$. The position vector of R and S will be, where RS is the shortest distance between the given lines is

(A) $\vec{a} + \dfrac{\left[(\vec{a}-\vec{b})\cdot(\vec{\beta}\cdot(\vec{\beta}\times\vec{\alpha}))\right]\cdot\vec{\alpha}}{|\alpha\times\beta|^2}$

(B) $\vec{a} + \dfrac{\left[(\vec{a}-\vec{b})\cdot(\vec{\beta}\times(\vec{\beta}\times\vec{\alpha}))\right]\vec{\alpha}}{|(\alpha\times\beta)|^2}$

(C) $\vec{b} + \dfrac{\left[(\vec{a}\times\vec{b})\cdot(\vec{\alpha}\times(\vec{\beta}\times\vec{\alpha}))\right]\vec{\beta}}{|\alpha\times\beta|^2}$

(D) $\vec{b} + \dfrac{(\vec{a}-\vec{b})\cdot(\vec{\alpha}\times(\vec{\beta}\times\vec{\alpha}))\vec{\beta}}{|\alpha\times\beta|^2}$

44. A and B are two events defined as follows:
A : It rains today with probability P(A) = 40%. **B :** It rains tomorrow with probability P(B) = 50%
Also, probability P (It rains today and tomorrow) = 30%. Also, $E_1 : P\left(\dfrac{(A\cap B)}{(A\cup B)}\right)$,

$E_2 : P\left(\dfrac{\left\{(A\cap\bar{B}) \text{ or } B\cup\bar{A}\right\}}{A\cup B}\right)$, then which of the following is/are TRUE?

(A) A and B are independent

(B) $P\left(\dfrac{A}{B}\right) < P\left(\dfrac{B}{A}\right)$

(C) E_1 and E_2 are equiprobable

(D) $P\left(\dfrac{A}{A\cup B}\right) = P\left(\dfrac{B}{A\cup B}\right)$

45. Let A_n be an $n \times n$ matrix in which diagonal elements are 1, 2, 3, 4, ... n
(i.e., $a_{11} = 1$, $a_{22} = 2$, $a_{33} = 3$, ... $a_{nn} = n$) and all other elements are equal to n
(A) A_n is singular for all n
(B) A_n is non-singular for all n
(C) det $A_n = (-1)^n \cdot n!$
(D) det $A_n = (-1)^{n+1} \cdot n!$

46. If $C_r = {}^nC_r$, then sum of the series $S = C_0^2 + \dfrac{C_1^2}{2} + \dfrac{C_2^2}{3} + ...$ upto (n + 1) terms, which of the following options is/are TRUE?

(A) $\dfrac{1}{(n+1)} \cdot {}^{2n+1}C_{n+1}$

(B) $\dfrac{1}{2(n+1)} \cdot {}^{2n+2}C_{n+1}$

(C) $\dfrac{1}{(n+1)} \cdot {}^{2n+1}C_n$

(D) $\dfrac{(2n+1)!}{((n+1)!)^2}$

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

47. Let $x_1 < x_2 < x_3$ be the three real roots of the equation $\sqrt{2014}\, x^3 - 4029x^2 + 2 = 0$, then $x_2(x_1 + x_3)$ is

48. There exists a unique strictly increasing sequence of non-negative integers $a_1 < a_2 < ..... < a_k$ such that $\dfrac{2^{289}+1}{2^{17}+1} = 2^{a_1} + 2^{a_2} + ... + 2^{a_k}$, then (k – 130) is

49. 10 chairs are arranged in a circle. Then the number of subset of this set of chairs that contain at least three adjacent chairs is N, then (N – 580) is

50. Let a_n and b_n are real number such that $(2+i)^n = a_n + b_n i$ for all integers $n \geq 0$ where $i = \sqrt{-1}$, if $\sum_{n=0}^{\infty} \frac{a_n b_n}{7^n}$ is k, then 16k is equal to

51. Let the sequence is 4, 7, 1, 8, 9, 7, 6, ... for n > 2 the nth term of the sequence is the units digit of the sum of the two previous terms. Let S_n denote the sum of the first n terms of this sequence. The smallest value of n for which $S_n > 10{,}000$ is p, then (p – 1990) is

52. Let ω and z be complex numbers such that $|\omega| = 1$ and $|z| = 10$. Let $\theta = \arg\left(\frac{\omega - z}{z}\right)$. Let $\frac{1}{m}$ is the maximum possible value of $\tan^2\theta$, then (m – 95) is

53. The number of rational number r, $0 < r < 1$, such that when r is written as a fraction in lowest terms the numerator and denominator have a sum of 1000 is N, then (N – 197) is

54. Let A_n and B_n be square matrices of order 3, which are defined as $A_n = \left[a_{ij}\right]$ and $B_n = \left[b_{ij}\right]$ where $a_{ij} = \frac{2i+j}{3^{2n}}$ and $b_{ij} = \frac{3i-j}{2^{2n}}$ $\forall$ i and j, $1 \leq i, j \leq 3$, if $\ell = \lim_{n\to\infty} T_r\left(3A_1 + 3^2A_2 + 3^3A_3 ..... + 3^nA_n\right)$ and $m = \lim_{n\to\infty} T_r\left(3B_1 + 2^2B_2 + 2^3B_3 ..... + 2^nB_n\right)$, then the value of $\frac{\ell + m}{3}$, where T_r (A) represent trace of matrix A